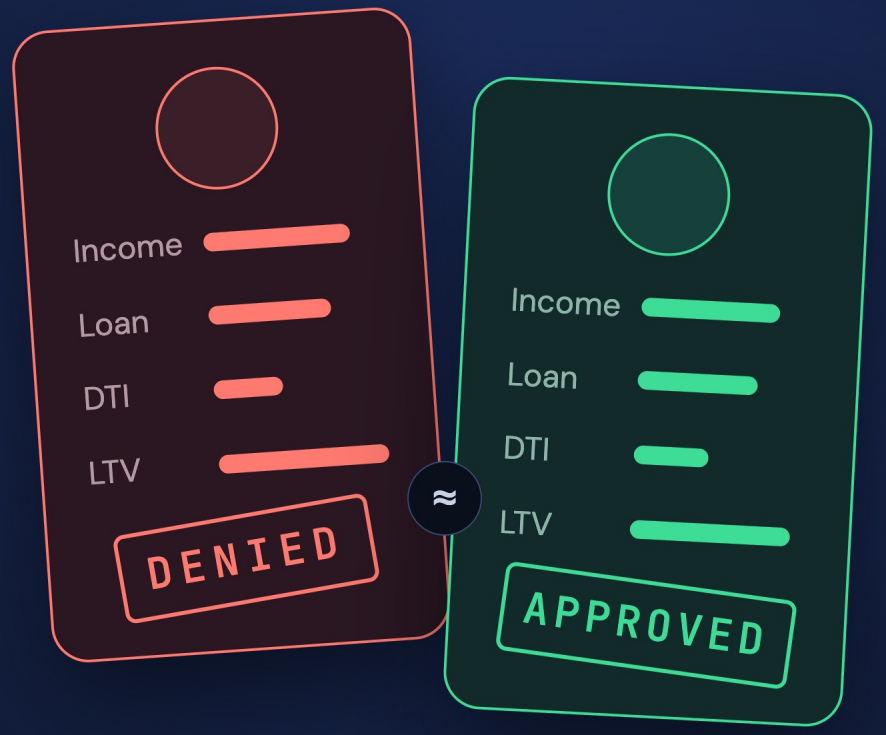
IBM Z DATATHON 2026 · REAL-TIME AI FOR CRITICAL DECISIONS

SecondLook

Same qualifications. Different decisions. Who notices?



TEAM WISE | AI FOR GOOD | ADVISORY ONLY

THE PROBLEM

Two applicants. Same profile. Different decisions.

Who checks why?

DENIED APPLICATION	
Income	$141K
Loan amount	$615K
Debt-to-income	46%
Loan-to-value	95%

DENIED

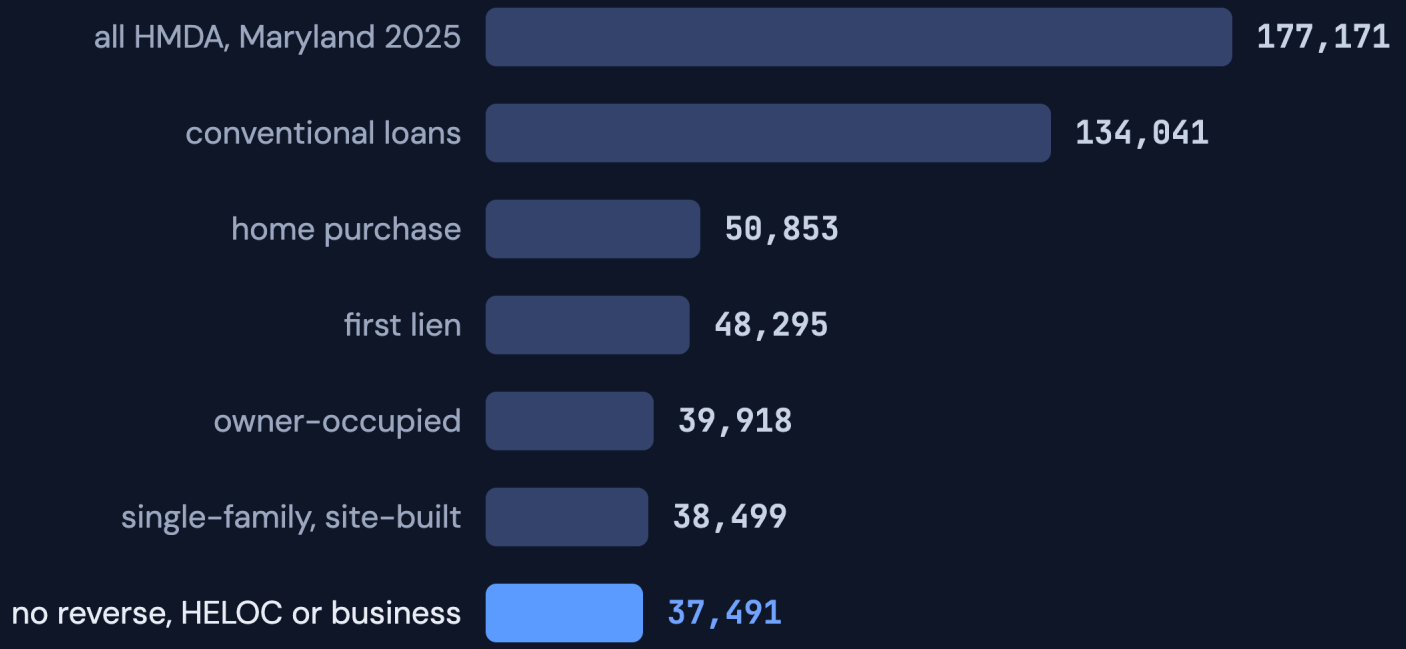
≈

APPROVED TWIN · SAME LENDER	
Income	$146K
Loan amount	$595K
Debt-to-income	45%
Loan-to-value	95%

APPROVED

THE DATA

From 177,171 records to 37,491 comparable applications



One product family, so every approved and denied application is judged on the same footing.

34,891
approved (originated or approved, not accepted)

2,600
denied

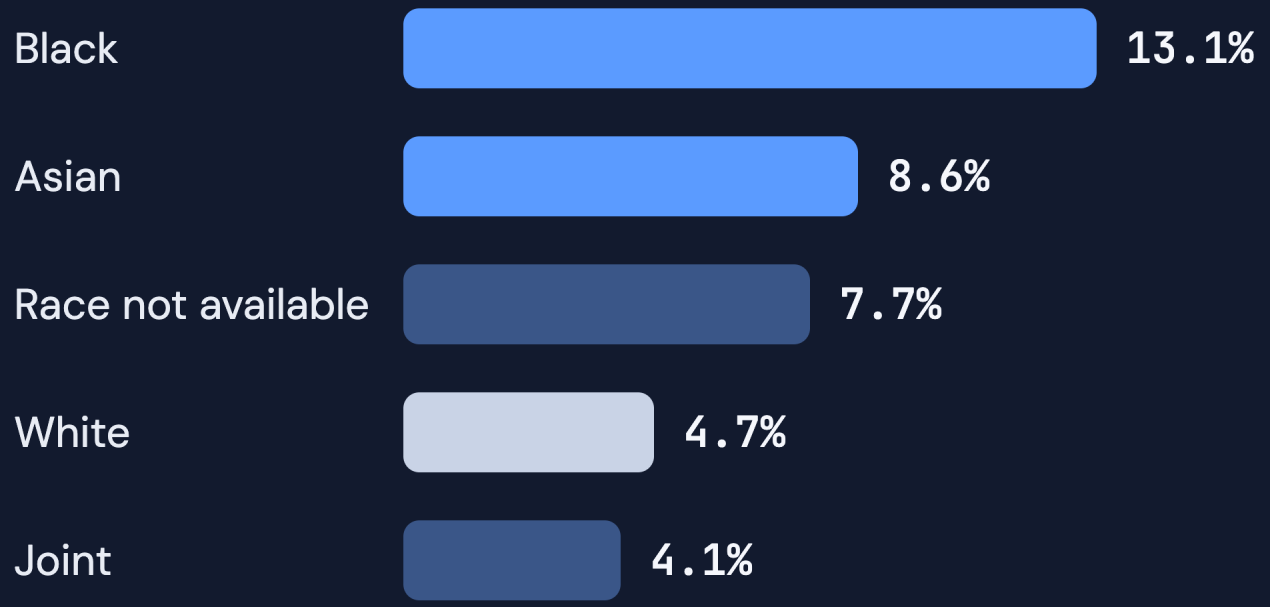
447
lenders

SOURCE: HMDA LAR 2025, FFIEC · MISSING: INCOME 0.8%, DTI 0.6%, LTV 1.0%

EXPLORING THE DATA · 1

Before matching, denial rates differ across groups

DENIAL RATE BY APPLICANT GROUP (RAW)



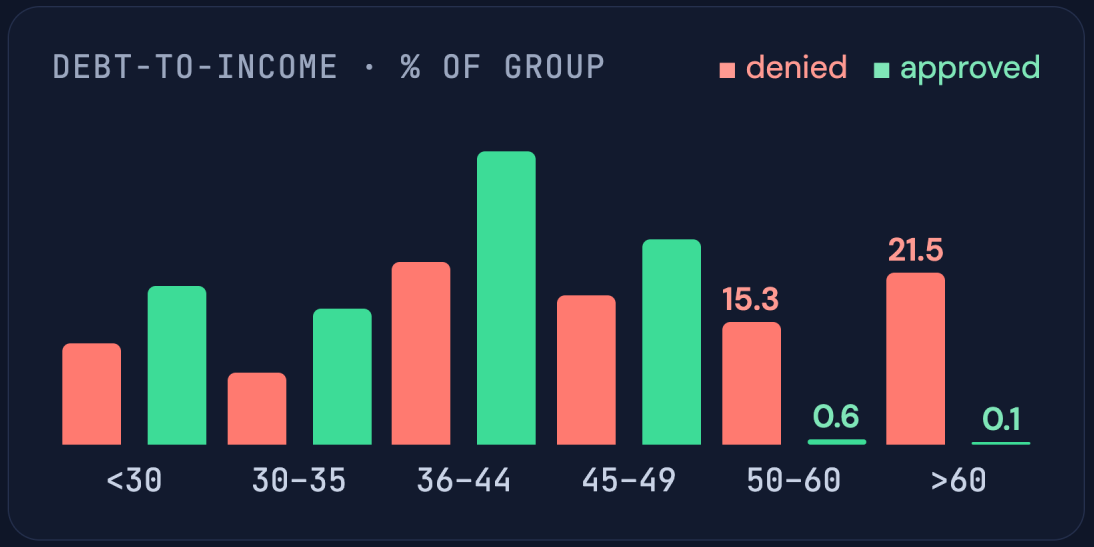
APPLICATIONS (n)	
Black	6,031
Asian	3,884
White	20,339
Joint	1,155
Race not available	5,802

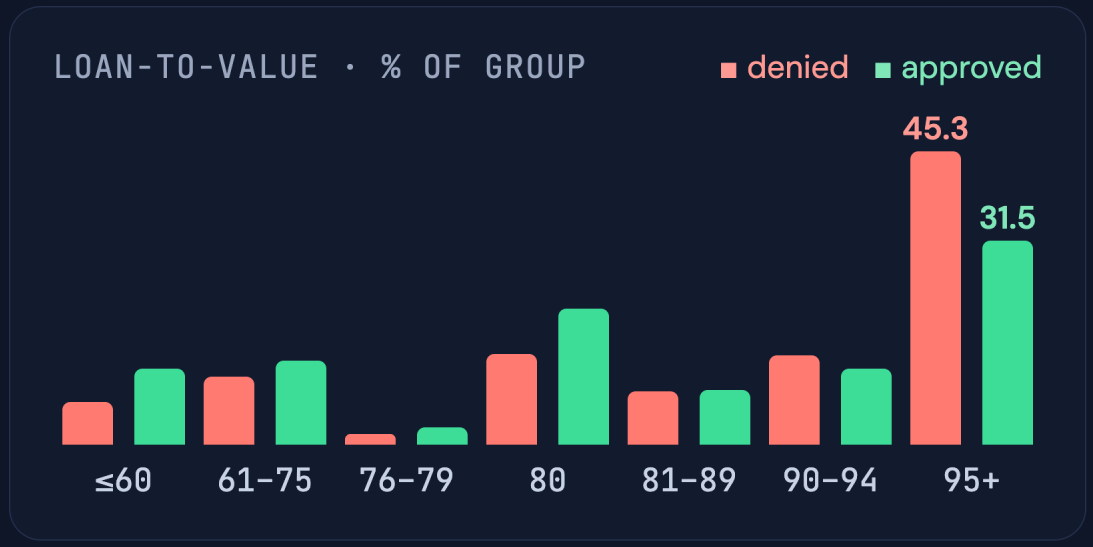
A raw gap is not an adjusted gap

Raw rates mix finances, lenders and places. So SecondLook matches on finances first, then compares.

COHORT OF 37,491 · GROUPS UNDER 150 APPLICATIONS SUPPRESSED

Finances overlap, except at very high DTI





35.6% of denials have DTI of 50 or more. Only 0.7% of approvals do.

Only 6.5% of denials with DTI above 60 have a comparable approval.

DTI ≥ 50 IS REPORTED AS A BRACKET IN HMDA · 2,600 DENIED, 34,891 APPROVED

THE SOLUTION

Match first. Explain second. A human decides.

01
Financial Twins Matching Engine
Find comparable applications

→

02
Structured Case Evidence
Financial differences, denial reasons, missing information

→

03 · LINUXONE
IBM Granite
Explains, then every sentence is checked against the facts

→

04
Human Reviewer
Verifies the evidence and decides next steps

STAGE 1 · MUST MATCH EXACTLY

- same lender
- conventional
- home purchase
- first lien
- owner-occupied
- site-built home

STAGE 2 · HOW CLOSE? (TOLERANCES)

- income ±15%
- loan ±10%
- property ±10%
- DTI ±5 pts
- LTV ±5 pts
- term ±60 mo

A similarity score, not a probability of approval.

Blind by construction

Race, sex, ethnicity and age live in a separate table, joined only after matching.

DENIAL REASONS ARE SHOWN TO REVIEWERS, NEVER USED FOR MATCHING

FINANCIAL TWINS · CASE b81d

One denied case, three approved twins

FIELD	DENIED	TWIN 1	TWIN 2	TWIN 3
Income ($K)	141	141	146	135
Loan amount	$615,000	$615,000	$595,000	$595,000
DTI (%)	46	46	45	43
LTV (%)	95	95	95	95
Property value	$645,000	$645,000	$625,000	$635,000
Distance	—	0.00	0.18	0.23
Match	—	STRONG	STRONG	STRONG
Outcome	DENIED	APPROVED	APPROVED	APPROVED

REASON CHECK

Stated reason: collateral

Applicant LTV 95 vs comparators' median 95. Not distinguished by reported data. Appraisal details are not public.

NOT IN PUBLIC DATA

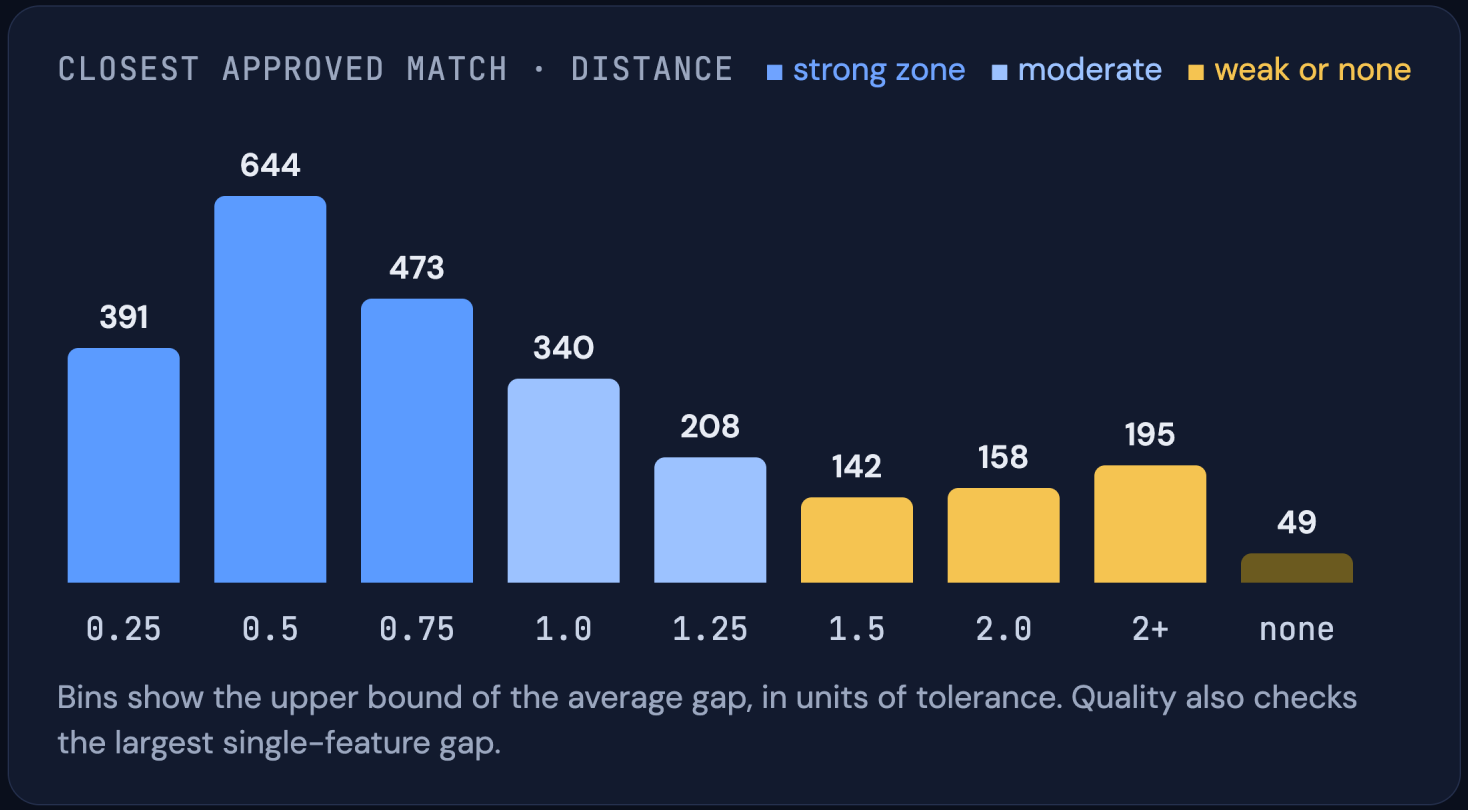
Credit score · assets · employment · documentation · lender policy overlays

So a difference in outcome is a question for review, not a verdict.

SAME LENDER · 1,915 APPROVED CANDIDATES · DISTANCE = MEAN GAP IN TOLERANCE UNITS

REVIEW QUEUE · 2,600 DENIALS

Not every denial is forced into the queue



Bins show the upper bound of the average gap, in units of tolerance. Quality also checks the largest single-feature gap.



874 · insufficient comparable evidence

506 of them have DTI above 60%, where almost no approved applications exist. We say so; we never force a flag.

EXPLAIN · ASK · ACT

AI assists. The reviewer stays in control.

1 · STATED REASON VS DATA

Collateral

Not distinguished by reported data

LTV 95 vs median 95 for approved twins

Every reason gets one of three labels:

✓ Consistent with reported data

≈ Not distinguished by reported data

? Not assessable with public data (credit history, employment)

2 · IBM GRANITE NOTE ✓ validated

"This denied application was queued because 3 approved applications from the same lender, out of 1,915 candidates, report closely similar income, loan amount, debt-to-income ratio, loan-to-value ratio, property value and loan term. The stated denial reason was collateral."

Granite sees facts only: no dollar amounts, no demographics. Invented numbers or accusing words are rejected.

3 · ASK, THEN DECIDE

What could explain the denial?

Answers are built from structured results only.

ⓘ Needs additional information

○ Reviewed, no conclusion

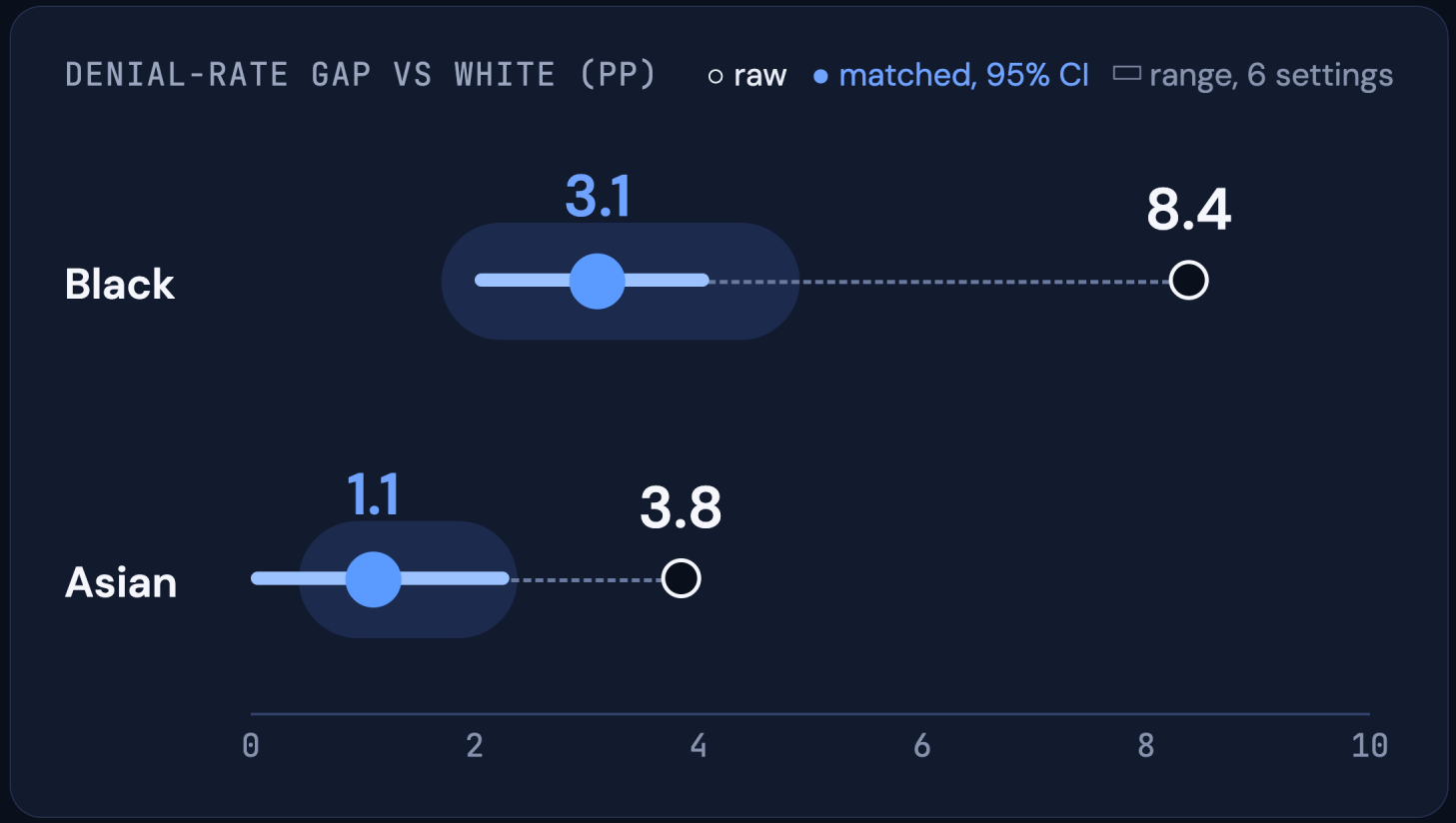
▲ **Escalate for further review**

Audit trail: reviewer-1 · v0.1 · strong match

🔒 **Advisory only. The API rejects any decision except these three (tested).**

BLIND REVIEW → REVEAL

Match on finances first, then reveal



An observed disparity. Not proof of discrimination.

Strong finance matches exist for 69% of Black and 59% of Asian applicants.

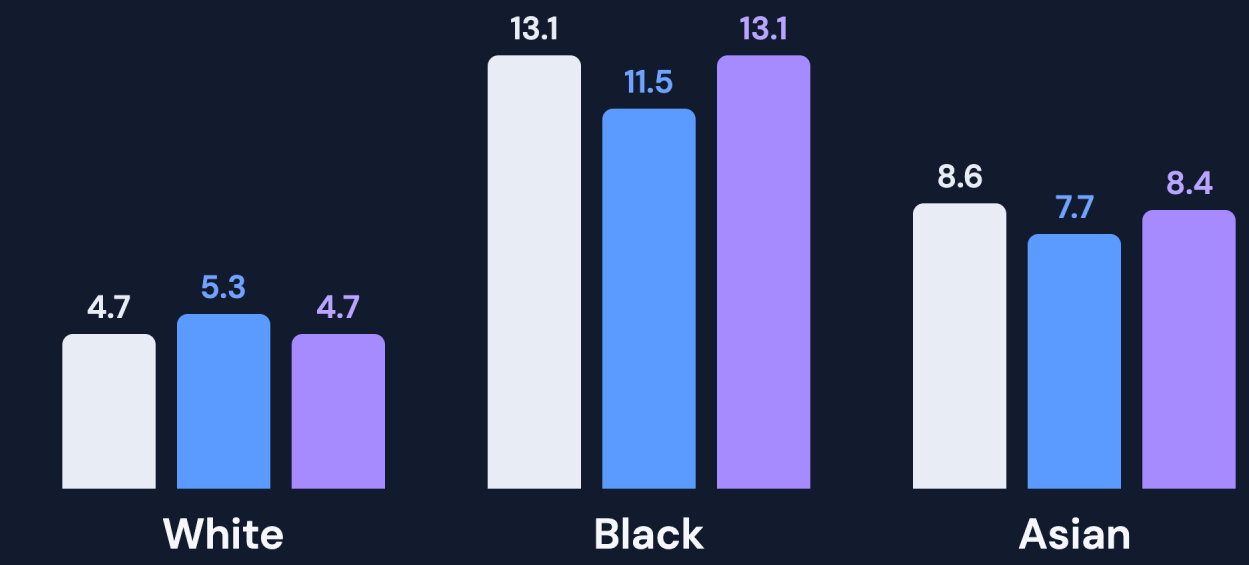
Black gap ranges 1.7 to 4.9 pp across settings. Asian interval touches zero.

Credit scores are missing. Finances and county partly predict race (AUC 0.68 and 0.74), so “blind” is not proxy-free.

MODEL SENSITIVITY ANALYSIS · XGBOOST ON LINUXONE

Do the public fields explain the gap? Not fully.

DENIAL RATE (%) · ACTUAL VS MODEL



0.867 → 0.873

AUC without vs with demographics. Overall accuracy barely moves.

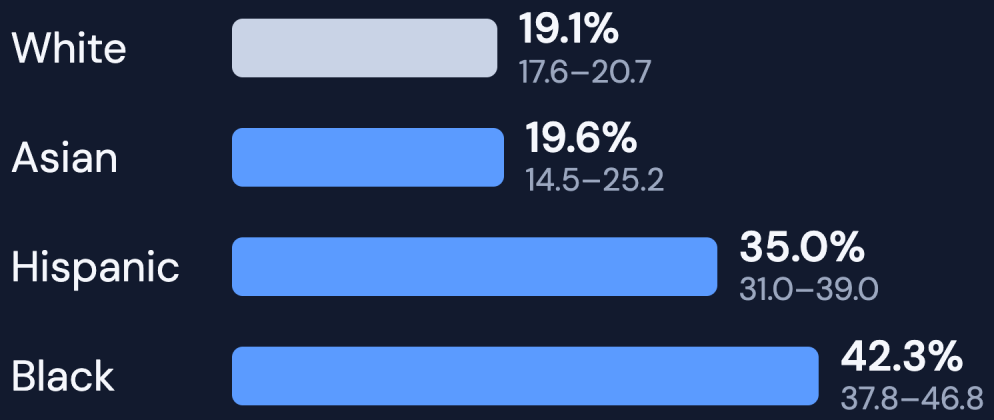
Without demographics the model under-predicts denials for Black applicants (11.5% vs 13.1%). The public finance, lender and county fields do not fully explain group differences.

Predicts historical decisions, not correct ones. Not evidence that a lender used race. The with-demographics model is for auditing only.

The pattern is not only in this dataset

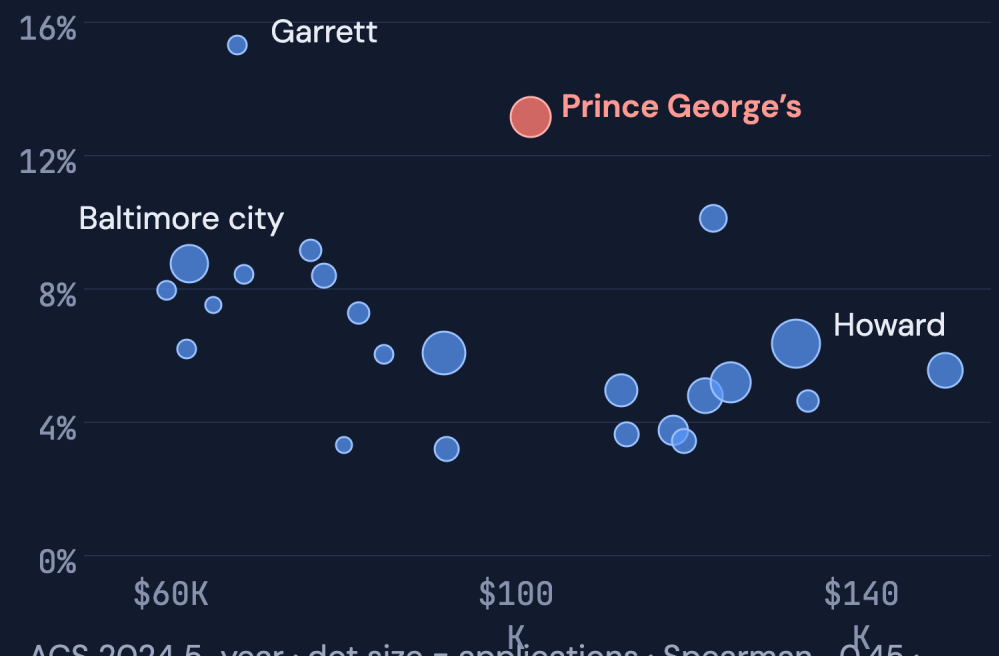
TURNED DOWN FOR CREDIT, PAST 12 MONTHS

US adults who applied for credit · Federal Reserve SHED 2025, survey-weighted



All credit types, not only mortgages. Never joined to HMDA applicants.

MARYLAND COUNTIES · INCOME VS DENIAL RATE



ACS 2024 5-year · dot size = applications · Spearman −0.45 · county level, not individuals

IBM LINUXONE · S390X · MEASURED, NOT CLAIMED

One controlled platform for a sensitive workflow

HMDA
SHED · ACS

LINUXONE · S390X · 3 vCPU · CONTAINER + SERVICES

Twin matching | Blind audit | XGBoost | Granite 4.0

Reviewer
SSH tunnel only

THROUGHPUT (REQ/S) VS CONCURRENT CLIENTS



~4 ms
median live twin search, simulated application

identical
result digests on s390x and on an arm64 laptop

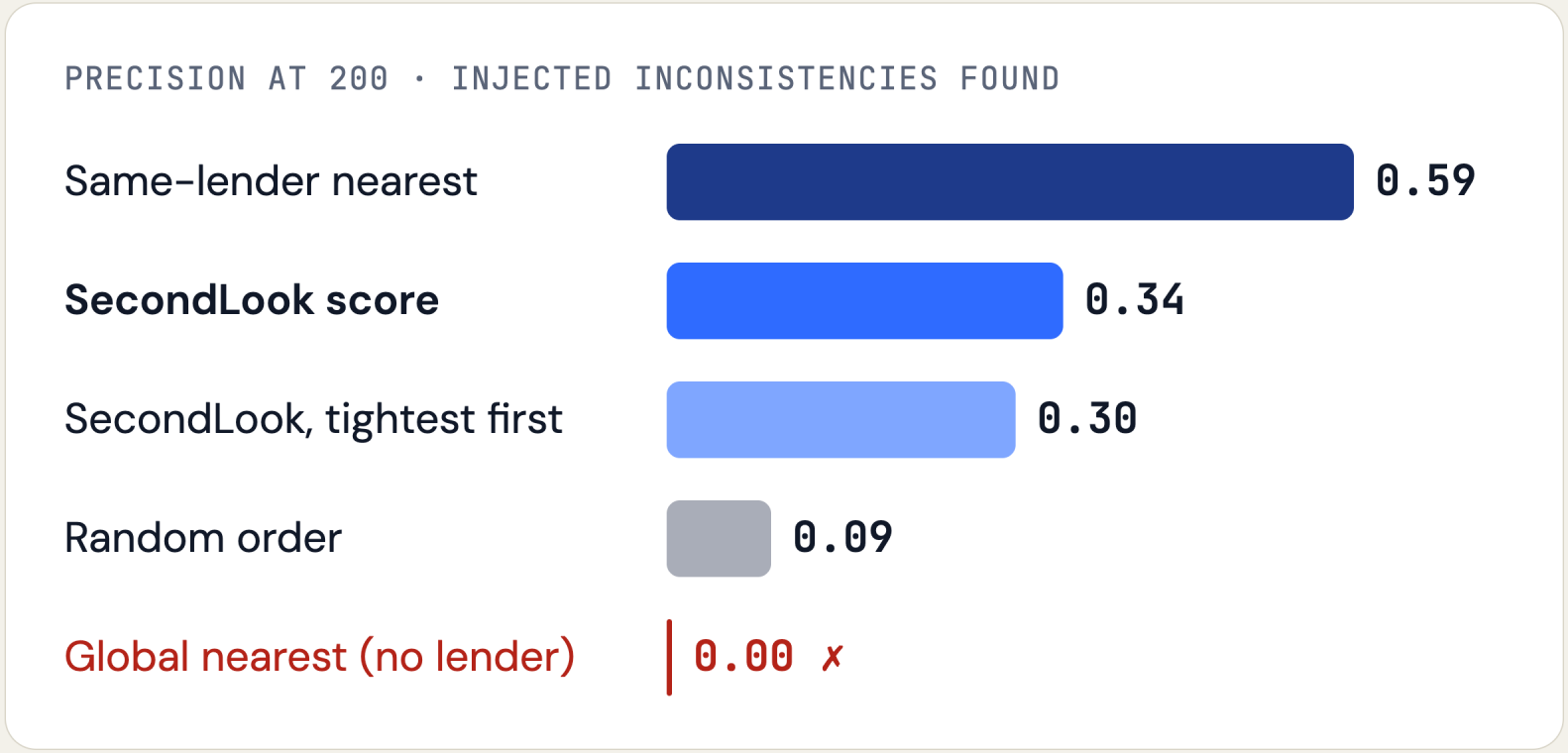
3 workers
~118 MB each, systemd services, auto-restart

token-gated
demographic endpoints; no personal data in logs

Straight talk: not faster than a laptop (0.68 s vs 0.38 s for same-lender matching). zDNN initialises, but offload was no faster, so we claim no acceleration.

VALIDATION · SYNTHETIC TEST

An honest test: the lender constraint matters



Same lender is essential

Plain similarity ranks as well. No accuracy win claimed.

We add evidence and workflow

Tolerances, missing-data handling, reason check, audit trail.

Not measured: review time and real workload savings. Real wrongful denials have no ground truth in public data.

SYNTHETIC: 200 INJECTED NEAR-COPIES AMONG 2,600 REAL DENIALS · MEAN OF 5 SEEDS

SECONDLOOK · TEAM WISE · IBM Z DATATHON 2026

SecondLook doesn’t decide who deserves a mortgage.

It makes sure decisions worth questioning don’t disappear into millions of records.

PUBLIC DATA

ADVISORY ONLY

RUNS ON LINUXONE

github.com/shreyvivek/WISE_Datathon